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Current Opinion in Genetics & Development

The stem cell zoo for comparative studies of developmental tempo☆

Jorge Lázaro[1,2,*], Jaroslaw Sochacki[1] and Miki Ebisuya[1,3,*]

The rate of development is highly variable across animal species. However, the mechanisms regulating developmental tempo have remained elusive due to difficulties in performing direct interspecies comparisons. Here, we discuss how pluripotent stem cell-based models of development can be used to investigate cell- and tissue-autonomous temporal processes. These systems enable quantitative comparisons of different animal species under similar experimental conditions. Moreover, the constantly growing stem cell zoo collection allows the extension of developmental studies to a great number of unconventional species. We argue that the stem cell zoo constitutes a powerful platform to perform comparative studies of developmental tempo, as well as to study other forms of biological time control such as species-specific lifespan, heart rate, and circadian clocks.

Addresses
[1] European Molecular Biology Laboratory (EMBL) Barcelona, Dr. Aiguader 88, 08003 Barcelona, Spain
[2] Collaboration for joint PhD degree between EMBL and Heidelberg University, Faculty of Biosciences, Heidelberg, Germany
[3] Cluster of Excellence Physics of Life, TU Dresden, Arnoldstraße 18, 01307 Dresden, Germany

Corresponding author: Ebisuya, Miki (miki.ebisuya@tu-dresden.de)
* Twitter account: @JorgeLazaroF, @EbisuyaMiki

Current Opinion in Genetics & Development 2024, **84**:102149

This review comes from a themed issue on **Developmental mechanisms, patterning and evolution (2024): Developmental Timing**

Edited by **James Briscoe** and **Miki Ebisuya**

For complete overview of the section, please refer to the article collection, "Developmental mechanisms, patterning and evolution (2024): Developmental Timing"

Available online 9 January 2024

https://doi.org/10.1016/j.gde.2023.102149

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Introduction

Spatiotemporal coordination of morphogenetic events is necessary for the correct unfolding of embryo development. A tight regulation of the time parameter is therefore essential during embryogenesis for controlling the size and cellular composition of tissues [1]. Embryos from different animal species, despite using conserved molecular mechanisms, display differences in their developmental tempo [2]. What regulates the rate of development across species? How is the species-specific tempo encoded in the genome? How did this trait evolve? All these remain open questions in the field.

The similarity between vertebrate embryos during the phylotypic stage has offered researchers an attractive system to tackle these questions [3]. During embryogenesis, vertebrates manifest a canonical body plan, with embryos of similar size and shape undergoing identical organogenetic events. However, the rate of development during embryogenesis remains species-specific, with human, mouse, chicken, and zebrafish finishing their organogenesis in approximately 60, 15, 10, and 3 days, respectively [4,5]. Direct interspecies comparisons of developmental tempo have remained challenging due to the different growing temperatures and uterine environments of individual animals. Only recently, the emergence of stem cell-based models of development has offered an unprecedented opportunity to investigate the regulation of developmental time across species [6]. *In vitro* differentiation of pluripotent stem cells (PSCs) can be used to recapitulate cell- or tissue-autonomous developmental processes and study them under similar experimental conditions. The use of stem cells also opens up the possibility to expand the developmental studies to unconventional animal species with interesting phenotypes whose embryos would be hard to obtain due to practical or ethical reasons. Additionally, stem cell systems are very amenable to genetic modification and highly quantitative measurements, allowing a finer dissection of the molecular players underlying developmental processes. In this review, we discuss the recent efforts in understanding the regulation of developmental tempo across species using stem cell models and the future of these *in vitro* systems.

Interspecies comparisons of developmental tempo using a zoo of stem cells

Among the various examples of time control observed in development, the cell-autonomous gene expression oscillations of the segmentation clock represent an ideal

☆ Given the role as Guest Editor, Miki Ebisuya had no involvement in the peer review of the article and has no access to information regarding its peer-review. Full responsibility for the editorial process of this article was delegated to James Briscoe.

system for comparing developmental tempo across species. The segmentation clock is active in the cells of the presomitic mesoderm (PSM) and controls the sequential formation of body segments along the vertebrate body axis [7]. This involves the molecular oscillations of the hairy and enhancer of split (HES) family of genes with a species-specific periodicity. *In vitro* modeling of the segmentation clock through the differentiation of PSCs into PSM cells has allowed the study of its temporal control in up to seven mammalian species, including mouse, rabbit, cattle, pig, rhinoceros, human, and marmoset [8–14]. This revealed that the period of the segmentation clock scales with the speed of HES7 biochemical reactions (e.g. protein degradation and intron delay) but not with the animal body weight, suggesting a general mechanism of tempo regulation that is uncoupled from a potential allometric dependency [8,9]. Alterations of the HES7 biochemical kinetics can also influence the dynamics of the segmentation clock *in vivo* [15–17]. Moreover, the HES7 biochemical reaction speeds do not scale with the cellular metabolic rate across species. This indicates that energy output cannot solely explain the species-specific biochemical kinetics [8]. However, perturbations of metabolic pathways in human PSM cells, specifically changes in the balance between the oxidixed and reduced forms of nicotinamide adenine dinucleotide (NAD+/NADH ratio), can slow down or accelerate the segmentation clock through changes in the translation rates, highlighting the importance of metabolism in the control of developmental time [18]. Changes in the glycolytic state have also been linked with alterations in PSM differentiation and segmentation *in vivo* [19–21]. How biochemical kinetics are established so precisely and scaled across species, as well as the role of metabolism in their regulation, remains to be determined. Future studies focusing on the genetic control of developmental tempo will help us understand how are these species-specific phenotypes determined at the cellular level.

Stem cell-based models have also been widely used as a system to study developmental timing of neurogenesis. The specification of neuronal subtypes displays a species-specific temporal progression that can be recapitulated *in vitro* [22]. For example, spinal cord motor neuron differentiation takes 3–4 days in mouse but 2 weeks in human [23]. Differentiation of PSCs into spinal cord motor neurons revealed that, similar to the segmentation clock, the 2.5-times temporal scaling between mouse and human could be explained by changes in the biochemical reaction rates across species [24]. Another example of interspecies differences in tempo can be found during corticogenesis [25]. Human cortical neurons derived from PSCs show a prolonged timing of maturation as compared with other species, even when transplanted into mouse brains. This points to a cell-intrinsic mechanism for these developmental programs [26–29]. Metabolism has been shown to be a key regulator of cortical neuron maturation timing, with stimulation of mitochondrial metabolism being able to accelerate human neuronal maturation [30]. Moreover, a recent study highlighted the role of epigenetic regulation in the control of mouse and human cortical neuron maturation [31]. It will be important to address how similar the regulatory principles controlling the timing of neurogenesis are across different tissues and timescales, days in the case of spinal cord development and months for corticogenesis. Interestingly, extrinsic factors have also been found to impact the timing of neuronal differentiation, as shown by the acceleration of human neurogenesis during interspecies codifferentiation of human and mouse PSCs [32]. Stem cell-derived organoids, with their three-dimensional structures comprising several cell types, can be a useful platform to address the role of extrinsic factors and cell–cell interactions on developmental tempo. Brain organoids generated from human and other primate PSCs have revealed interspecies differences in the composition and rate of neuronal development [33,34]. Similarly, self-organization of optic cup structures in retinal organoids takes longer and forms bigger structures in human compared with mouse [35]. The use of organoid models will allow researchers to understand how the cell-autonomous developmental tempo can be integrated at a tissue level and regulate species-specific morphogenesis.

The delay in development caused by responses to environmental stress, known as diapause, has recently emerged as a potential system to investigate the regulation of developmental tempo. This physiological process allows animals to reversibly slow down and pause their developmental progression in order to optimize the timing of birth [36]. Diapause has been observed naturally in numerous species, and can be recapitulated *in vitro* using preimplantation embryos as well as stem cell models [37–39]. Inhibition of the mammalian target of rapamycin (mTOR) pathway can reversibly induce pausing of mouse blastocysts, human stem cell-based blastocyst models (i.e. blastoids), and human and mouse PSCs [39,40]. Interestingly, the speed at which the PSCs transition into a diapause-like state and some of the pathways involved in the maintenance of dormancy are different in mouse and human [39]. The mechanisms used by different species to regulate the entry and exit of diapause will be of great interest to uncover the key players involved in the control of developmental tempo.

Exploring biological time: expansion of the zoo and its clocks

Most studies of developmental tempo involving stem cell models have focused on comparing mouse and human. This is mostly because these two species show

large phenotypic differences and their stem cells have a great number of resources available for culture and differentiation. However, the study of other unconventional species is critical to assess the generality of the findings, determine what constitutes a species-specific behavior, and obtain more information about the evolutionary history of these developmental traits. For example, recapitulating *in vitro* the segmentation clock of six mammalian species has allowed the establishment of a general scaling law between the segmentation clock period and the speed of biochemical reactions [8]. Similarly, comparisons of human and primate brain organoids have revealed differences in the duration of particular developmental states that could explain the size expansion observed in human brains [33].

Reprogramming of adult somatic cells into induced pluripotent stem cells (iPSCs) has further enhanced our capacity to perform interspecies comparisons. Similar to embryonic stem cells (ESCs), iPSCs can proliferate indefinitely and differentiate into the three germ layers [41]. However, unlike ESCs, iPSCs are not derived from embryonic tissue, constituting a more practical stem cell source for many animal species. Over the years, a great amount of iPSC lines have been derived from wild and farm animals, generating a zoo of stem cells across the globe [42]. In Supplementary Table 1, we made a list of animal species with available PSCs, including bat, rhinoceros, seal, opossum, as well as several primates. The unique development of some of these species will allow the study of particular aspects of developmental time. For example, marsupial models of development could be useful to investigate the emergence of tissue-specific changes in developmental tempo, known as heterochronies. Marsupial embryogenesis is characterized by the faster development of the upper body as compared with the lower body, enabling them to climb into the pouch after birth and complete development [43,44]. *In vitro* modeling of embryonic structures with different anterior–posterior identities could offer valuable insight into the molecular basis of these differences in developmental tempo. It is important to point out that most PSC lines to date have been derived from mammals. Despite the efforts in obtaining stable stem cells from other vertebrate species, a better understanding of the mechanisms that allow the generation and maintenance of mammalian PSCs is necessary to derive high-quality stem cells from birds, reptiles, amphibians, and fishes [45,46]. Some of the currently available nonmammalian PSCs can only be considered as partially reprogrammed. Moreover, quantitative studies of developmental kinetics in cells from ectothermic animals will require the normalization of the different temperature conditions.

Apart from extending our current investigations to other animals, the use of PSCs from unconventional species could enable the study of other forms of biological time. One example is species-specific lifespan. Species with a long lifespan tend to have larger size and longer gestational periods [47]. Interestingly, slower biochemical reaction rates such as protein turnover or transcription elongation speed have been associated with increased longevity [48,49]. These findings are consistent with the role of biochemical kinetics as regulators of developmental tempo. Species-specific mechanisms for extended lifespan are also reported, including the uniquely high efficiency and accuracy in DNA repair of the long-lived bowhead whale [50]. This is in line with the observations that somatic mutation rates are inversely correlated with lifespan across species [51]. Moreover, a pan-mammalian epigenetic signature has been seen to connect the methylation level of several developmental genes with the species' maximum lifespan, suggesting a relation between aging and development [52,53]. Most of these studies have been done using adult tissue samples, allowing researchers to harvest cells form unconventional species in a relatively simple way. However, adult cells are very limited in their capacity to proliferate, making it challenging to keep them in culture and perform experimental perturbations. Stem cell systems provide an unlimited source of material, amenable to genetic modifications and with the capacity to differentiate to various tissues. The use of stem cells from uniquely long- or short-lived species could allow researchers to address whether lifespan signatures are present during embryonic development, as well as how they are initially established and maintained.

Another parameter correlated with life expectancy is the heart rate. Among mammals, with the exception of the human species, there is an inverse correlation between heart rate and lifespan. Animals that live longer tend to have slower heart beats, with a remarkably constant number of heart beats per lifetime across animals [54]. Similarly, larger animals tend to have slower heart rates as compared with animals of smaller size [55]. The heart beat is initiated by the pacemaker cells located in the sinoatrial node, and its frequency can be modulated through various neurotransmitters [56]. However, how the species-specific basal heart rate is first established remains to be determined. The strong correlation with body weight suggests a connection between heart rate and the organismal metabolic rate as a biophysical imperative. Nevertheless, a cell-autonomous mechanism for establishing basal heart rates in pacemaker cells could also be possible. Recent protocols have allowed the differentiation of PSCs into cardiac pacemaker cells [57–59]. The study of these cells will enable researchers to investigate how species-specific heart rates are determined and what the influence of metabolism is on the beating frequency of cells and tissues.

PSCs could also help in the understanding of one of the most universal forms of biological time control found in

nature: the circadian clock. This clock is an endogenous molecular feedback loop with a period of approximately 24 h that allows organisms to anticipate daily changes in the environment caused by earth's rotation. The master circadian clock is located within the hypothalamus, but most cells in the body show circadian molecular oscillations [60]. The circadian clock appears during development, with PSCs having no apparent circadian oscillations [61]. Interestingly, the onset timing of the circadian clock appears to correlate with the rate of development, with rapidly developing species showing an earlier emergence of the clock [62]. *In vitro* models of PSC differentiation are already being used to understand the mechanisms regulating circadian clock appearance during development [61,63]. Furthermore, using PSCs from species living in polar environments — where regular photoperiods are largely absent — may provide valuable insights into the mechanisms allowing environment adaptation. Some of these species, such as arctic ground squirrel, have maintained their circadian clocks, while others, including reindeer, have lost their daily rhythmic activities [64–66].

Overall, the stem cell zoo offers a great platform to perform comparative studies of developmental processes. However, while PSC-derived models can recapitulate cell-intrinsic behaviors, they lack the ability to reproduce the complex environment and morphogenesis of the embryo. Xenotransplantation, species hybridization, or interspecies chimerism are useful approaches to address the importance of extrinsic factors in the establishment of species-specific behaviors *in vivo* [26,67,68]. The combination of the stem cell zoo with embryological data is essential to avoid *in vitro* artifacts and have a comprehensive understanding of how developmental tempo emerges across species.

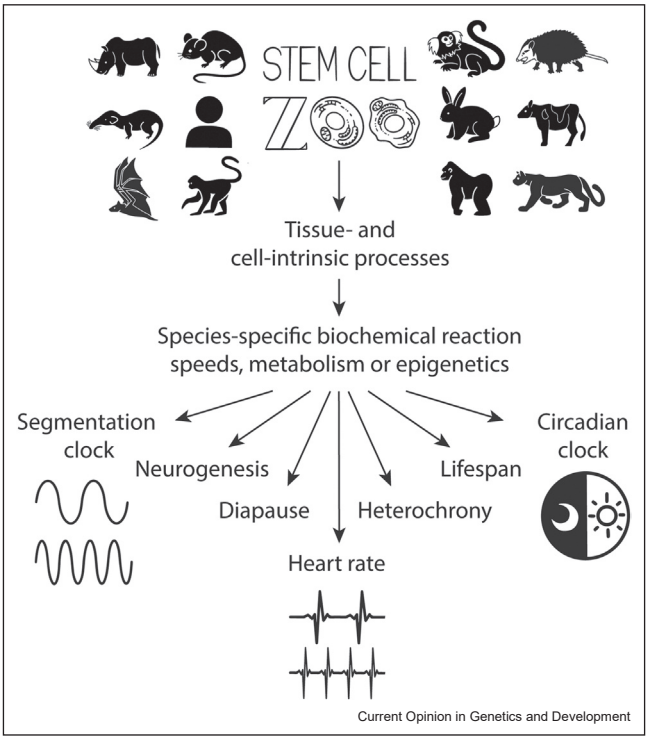
Summary

The use of *in vitro* models of development has advanced our understanding of the rules governing developmental tempo. Stem cell differentiation allows us to compare different species under the same experimental conditions and learn about the mechanisms that make animal species unique at the cellular and tissue level (Figure 1). The addition of more species to the stem cell zoo will facilitate further interspecies comparisons at many biological levels.

Funding

This work was supported by the European Molecular Biology Laboratory (EMBL), the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy — EXC 2068 — 390729961 — Cluster of Excellence Physics of Life of TU Dresden, the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (grant agreement No. 101002564) (to M.E.), and the Boehringer Ingelheim Fonds (BIF) PhD fellowship (to J.L.). M.E. is supported by the Alexander von Humboldt Foundation in the framework of the Alexander von Humboldt Professorship endowed by the Federal Ministry of Education and Research.

Figure 1



The stem cell zoo can be used to study temporal processes across species. PSCs from different animal species can be used to recapitulate tissue- and cell-autonomous mechanisms. Through changes in the speed of biochemical reactions, cellular metabolic rates, or epigenetics, PSC-derived models can effectively recapitulate species-specific differences across several biological processes.

CRediT authorship contribution statement

J.L. and M.E. conceived and wrote the paper. J.L. and J.S. assembled the list of animal species with available PSCs.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could

have appeared to influence the work reported in this paper.

Acknowledgements

We are thankful to Júlia Charles Aymamí for designing the animal icons and illustrations of the figure, to Maria Costanzo, Kristina Stapornwongkul, Guillermo Martínez Ara, and Abiram Eliab Olivares Resendiz for providing feedback.

Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.gde.2023.102149.

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